

Vectors



Vector arithmetic in three dimensions

- 1 Given $\mathbf{a} = (2, -1, 3)$ and $\mathbf{b} = (1, 4, -2)$, find $\mathbf{a} + 2\mathbf{b}$ and $|\mathbf{a}|$.
 - 2 Find the magnitude of $\mathbf{v} = (6, -3, 2)$, and hence find a unit vector in the same direction.
 - 3 Find $2\mathbf{a} - \mathbf{b}$ for $\mathbf{a} = (1, 3, -2)$ and $\mathbf{b} = (4, -1, 2)$.
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The scalar (dot) product and the angle between vectors

- 4 Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 1, -2)$ and $\mathbf{b} = (-1, 4, 2)$, and state whether $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.
 - 5 Find the angle between $\mathbf{a} = (1, 2, 2)$ and $\mathbf{b} = (2, -1, 2)$, to the nearest degree.
 - 6 Find the value of k such that $\mathbf{a} = (3, k, -2)$ and $\mathbf{b} = (1, -2, 4)$ are perpendicular.
 - 7 Find the angle between $\mathbf{a} = (3, 0, 4)$ and $\mathbf{b} = (0, 5, 0)$, to the nearest degree.
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The vector equation of a line

- 8 Write the vector equation of the line through $(1, 2, -1)$ with direction vector $(2, -1, 3)$.
 - 9 Find the vector equation of the line through the points $A(2, 0, 1)$ and $B(5, 3, -2)$.
 - 10 Find the vector equation of the line through $(0, 1, -2)$ parallel to the line $\mathbf{r} = (3, 0, 1) + t(2, -1, 4)$.
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Classifying lines: parallel, perpendicular and collinear points

- 11 Find the acute angle between lines with direction vectors $(1, 2, 2)$ and $(2, 1, -2)$, to the nearest degree.
 - 12 Determine whether the lines $\mathbf{r}_1 = (1, 0, 2) + s(1, 1, 1)$ and $\mathbf{r}_2 = (2, 1, 0) + t(2, 2, 2)$ are parallel, and if so, whether they are the same line.
 - 13 Show that the points $A(1, 2, 3)$, $B(3, 6, 7)$, and $C(5, 10, 11)$ are collinear.
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Intersecting and skew lines

- 14 Determine whether the lines $\mathbf{r}_1 = (2, 1, 3) + s(1, 0, -1)$ and $\mathbf{r}_2 = (0, 2, 5) + t(1, 1, -2)$ intersect.
- 15 Find the point of intersection of the lines $\mathbf{r}_1 = (2, 1, 1) + s(1, 1, 2)$ and $\mathbf{r}_2 = (0, 5, 3) + t(2, -1, 1)$.
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The equation of a plane

- 16 Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.
- 17 Find the Cartesian equation of the plane containing the points $A(2, 0, 0)$, $B(0, 3, 0)$, and $C(0, 0, 4)$.
- 18 Find the Cartesian equation of the plane through $(1, -1, 2)$ perpendicular to the vector $(3, 0, -2)$.
- 19 Find the Cartesian equation of the plane through the points $A(2, 1, 0)$, $B(0, 2, 1)$, and $C(1, 0, 3)$.
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Angles between lines, planes, and pairs of planes

- 20 Find the angle between the line with direction $(2, 1, 2)$ and the plane $2x - y + 2z = 6$, to the nearest degree.
- 21 Find the angle between the planes $x + 2y - z = 3$ and $2x - y + z = 5$, to the nearest degree.
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Distances and same-side reasoning

- 22 Find the shortest distance from the point $(3, 1, -2)$ to the plane $2x - y + 2z = 6$, using
$$d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}.$$
- 23 Find the foot of the perpendicular from $P(4, 0, 1)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$, and the shortest distance from P to the line.
- 24 Determine whether the point $(1, 2, 3)$ lies on the same side of the plane $x + 2y - z = 4$ as the origin.
- 25 Find the distance from the origin to the plane $3x + 4y = 20$ (which does not depend on z).
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A well-designed word problem: a drone's flight path

- 26** This question has four parts. A drone's position at time t seconds ($t \geq 0$) is $\mathbf{r}(t) = (2, 0, 10) + t(3, 4, -1)$, in metres, with z measured vertically. (a) Find the drone's position after 4 seconds. (b) The ground is the plane $z = 0$. Find the time at which the drone lands. (c) Find the drone's horizontal distance from the origin (in the xy -plane) at the moment it lands. (d) Find the angle between the drone's flight path and the ground, to the nearest degree.